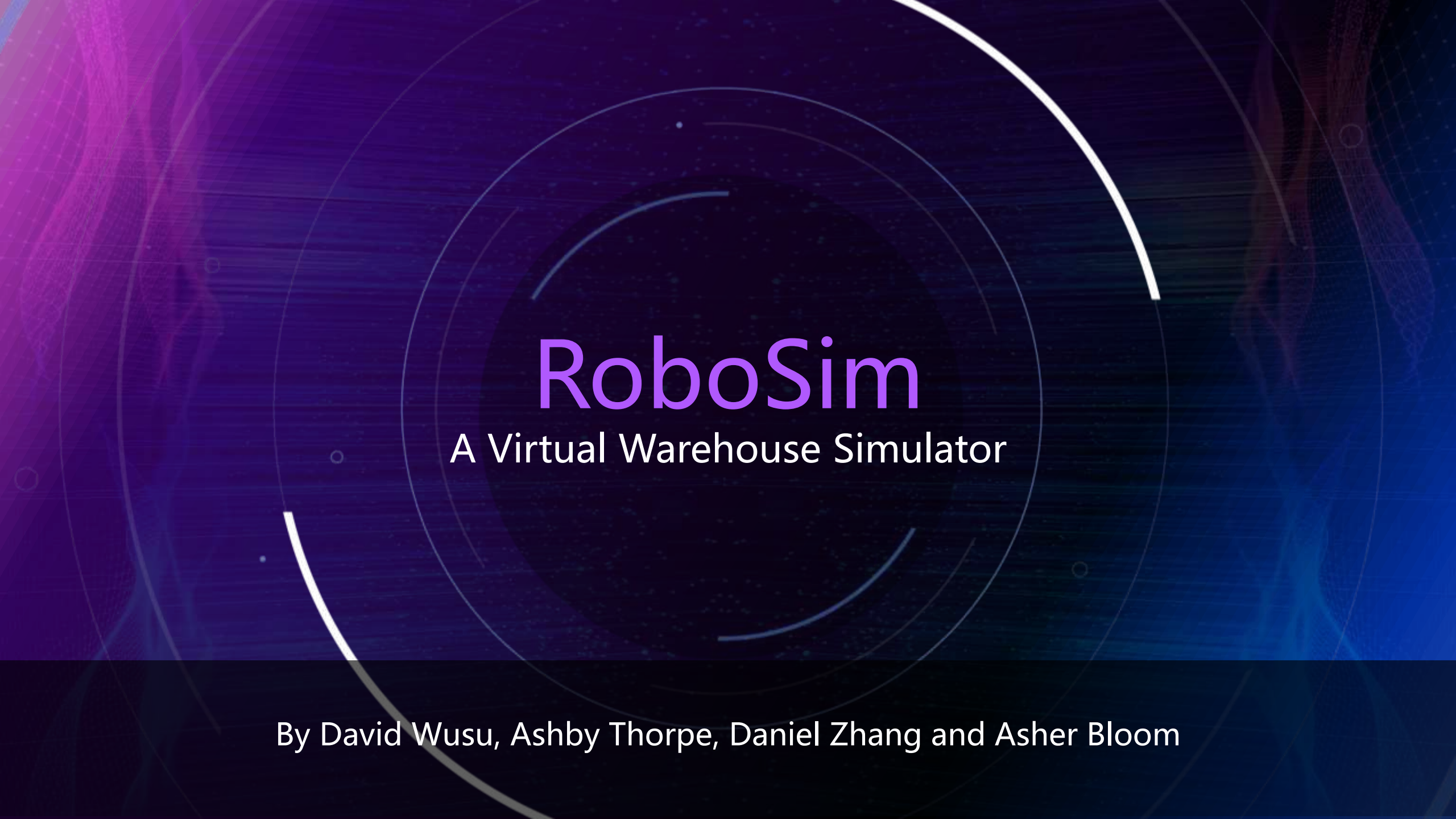


The background features a dark blue to purple gradient. Overlaid on this are several concentric circles of varying radii, some of which are partially cut off by the edges of the frame. A fine grid pattern is visible in the background, particularly on the left side where it transitions into a magenta/purple hue. The text is centered within the composition.

RoboSim

A Virtual Warehouse Simulator

By David Wusu, Ashby Thorpe, Daniel Zhang and Asher Bloom

Our Project

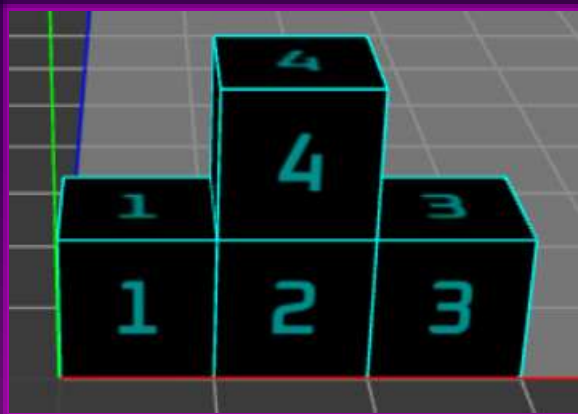
- Client: Marius Jurt – Engineer at Toshiba
- Business Problem: Explore a new type of robotic box that could revolutionise warehouse management with self-moving and self-stacking capabilities
- Objective: Improve the existing platform to do the following:
 - Allow 3D routing algorithms to be benchmarked
 - Enable experimentation of box configurations



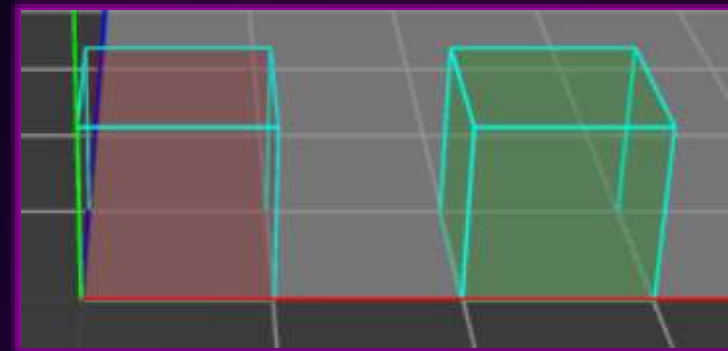
Terminology Preface

- **Scene:** A given arrangement of boxes
- **Scenario:**
 - Consists of a **start scene** and an **end scene**
 - **End scene** is **start scene** with some boxes moved to **different** locations
- **Algorithm:** Python program that aims to **complete** a **scenario**

Scene



Scenario



User Stories

Warehouse Managers

As a **Warehouse Manager**, I want to **optimise** storage and retrieval of goods to **minimise** management overheads

Algorithm Designers

As an **Algorithm Designer**, I want to **develop** and **benchmark** algorithms to **streamline** optimisation of 3D robot navigation

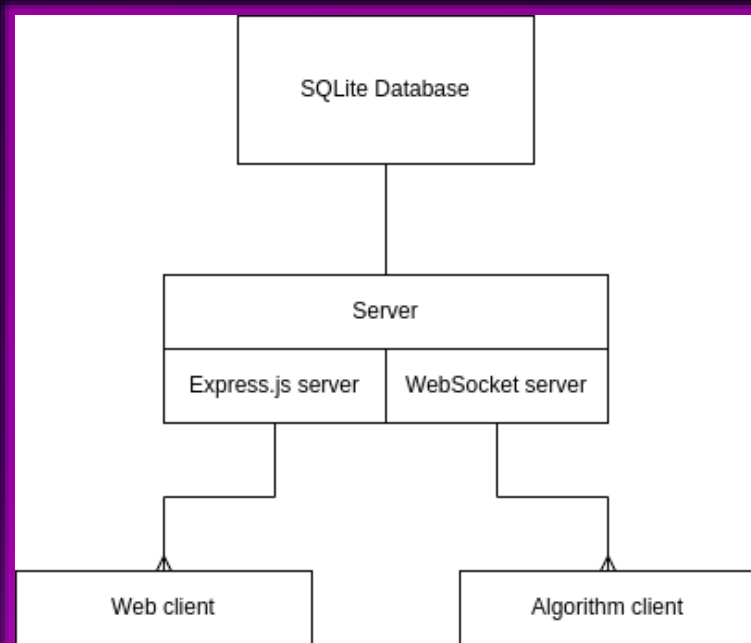
Toshiba Developers

As a **Toshiba Developer**, I want to **explore** methods of storing goods in 3D space to **improve** efficiency of clients' business functions

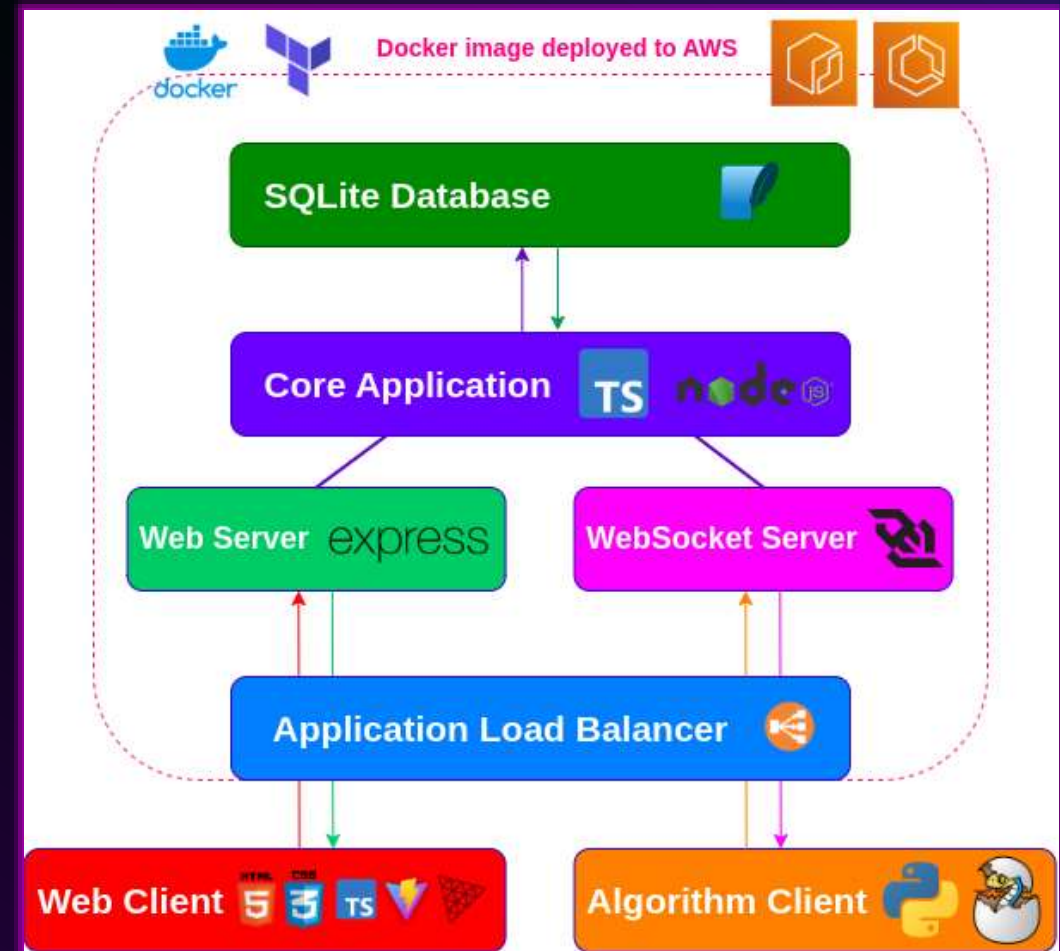
Demonstration (Old Project)

Project Architecture & Tech Stack

MVP/Beta



Final



Code Refactors

- Refactored the entire server to improve code simplicity and maintainability
- Allowed multiple algorithms to connect simultaneously
- Ran algorithms on a fixed sequence of scenarios
- Changed Python project to a Hatch project
- Removed restrictions on when moves can be sent
- Cleaned up frontend code

API Improvements

○ Before

```
moves = [...]  
i = 0  
  
def next_move(server: Server) → Moves:  
    global i  
    i += 1  
    if i < len(moves):  
        return moves[i]
```

● After

```
moves = [...]  
  
async def algorithm(client: Client, scenario: Scenario) → None:  
    for move in moves:  
        pending_move = await client.send_move(move)  
        await pending_move.accepted()
```

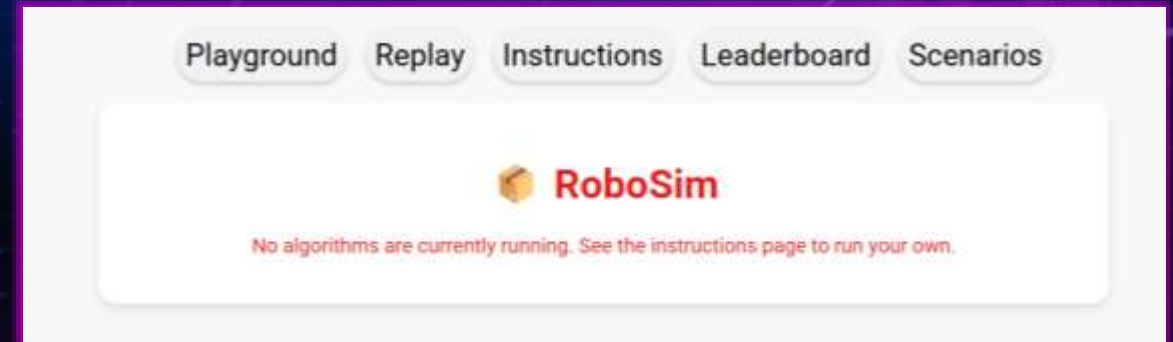
- Simplified the API of the Python package
- Published comprehensive documentation on GitHub Pages

IDP 1: UI

Step 1 – New Colour Scheme & Shared Navbar

- New colour scheme to fit **robotic** theme
- Shared header for **easier** navigation

Before



After



Step 2 – Dynamic Main Page

- Created **dynamic** landing webpage **at client's request**
- Features 100 boxes moving **randomly**



Step 3 – New Box Texture

- Created new **robotic** texture for cubes
- Added **box IDs** to sides of boxes **at client's request**

Before

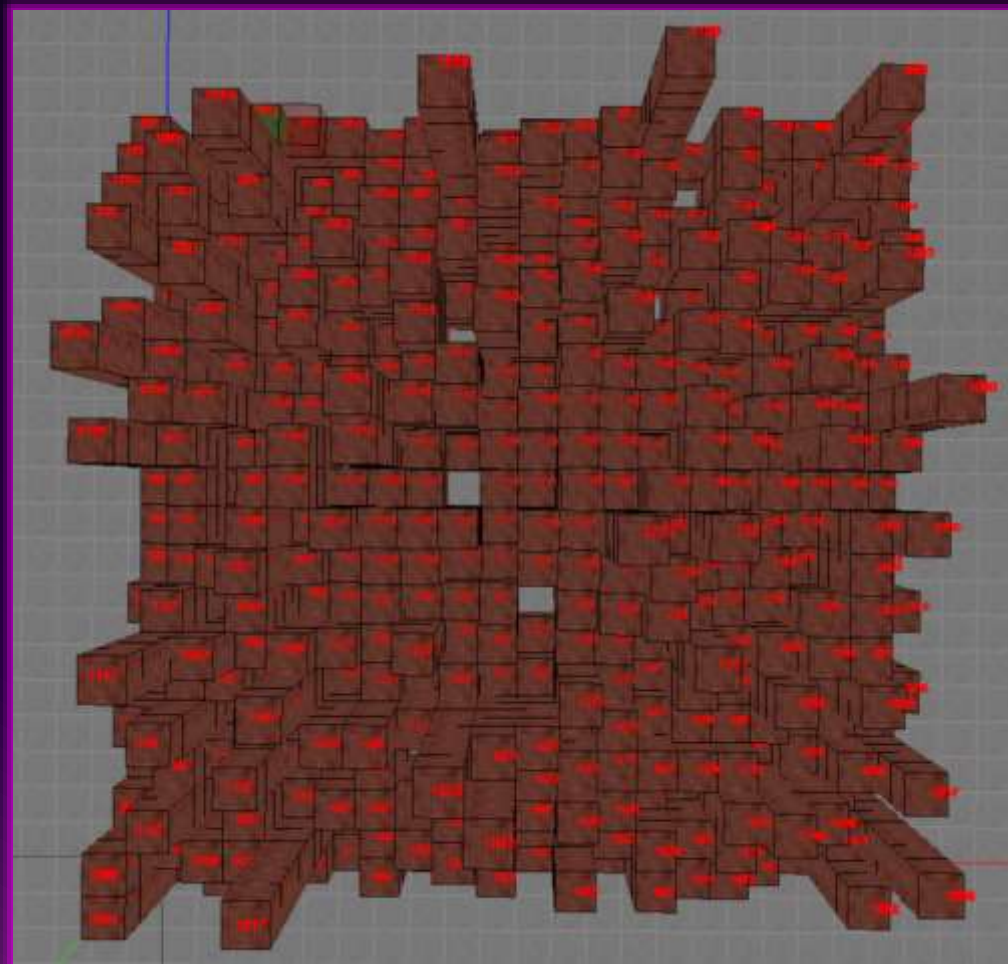


After

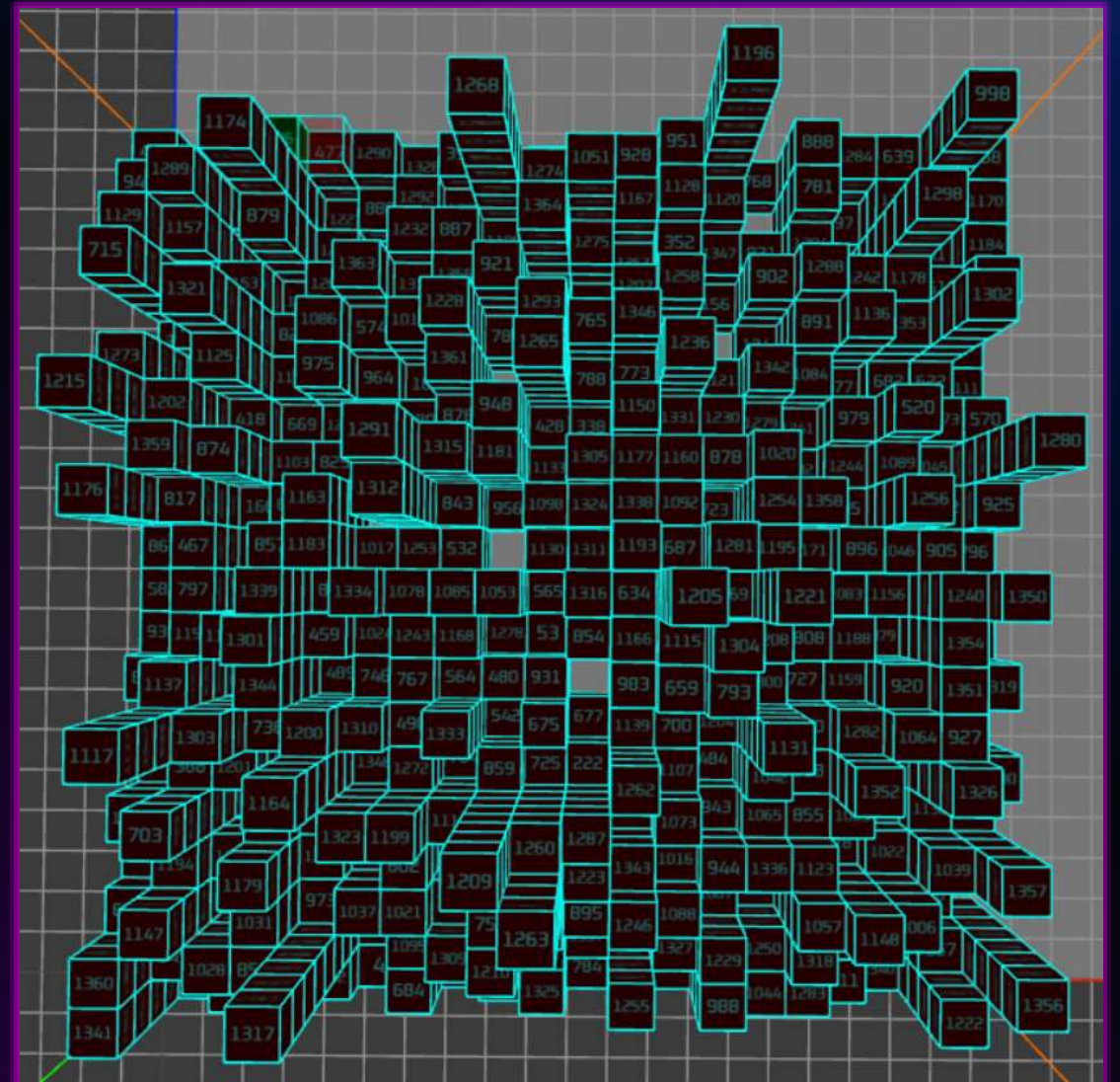


Step 3 – New Box Texture

Before



After



Step 4 - Dynamism

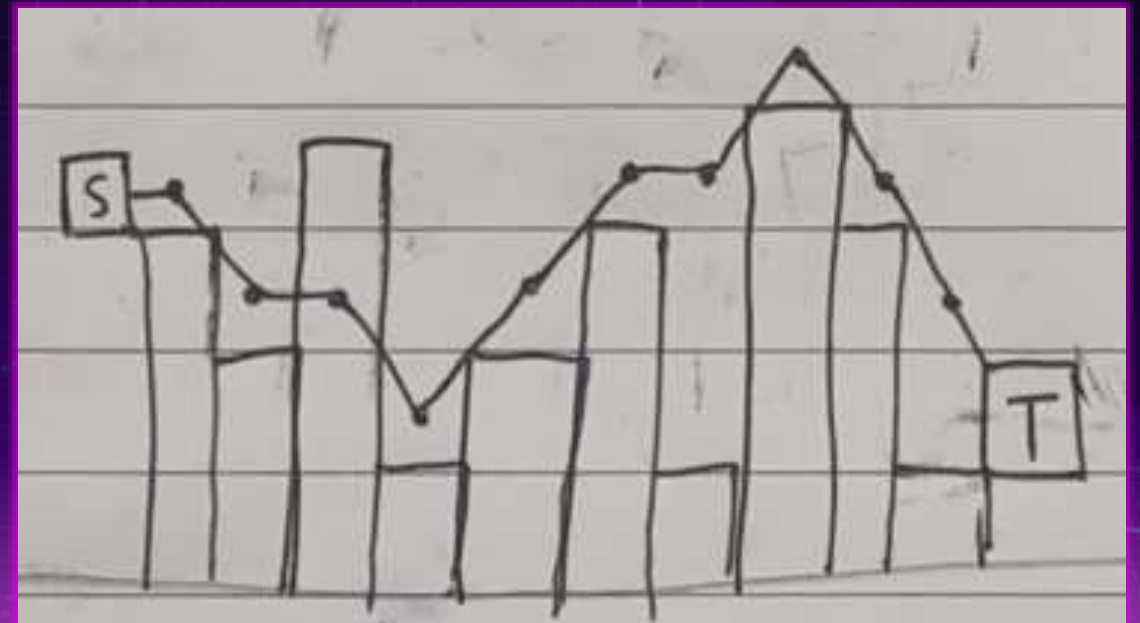
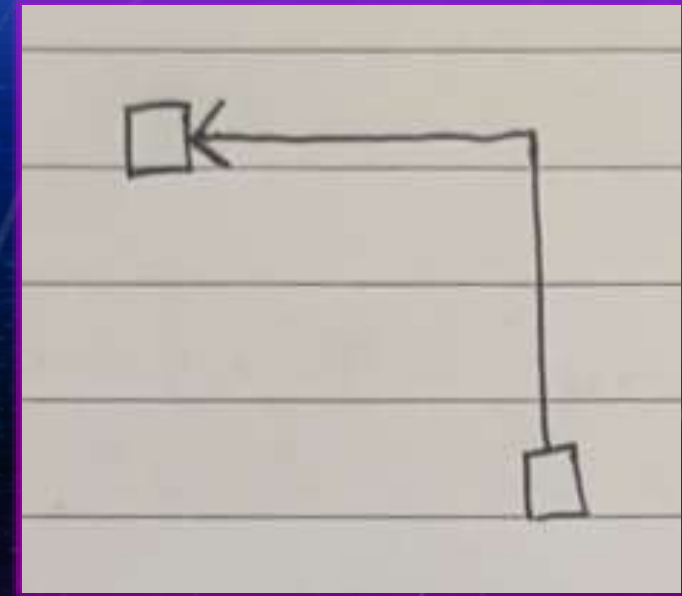


Result: Intuitive and appealing UI!

IDP 2: Fill-Remove (FR) Algorithm & Benchmarking

Step 1 - Fill-Remove Algorithm

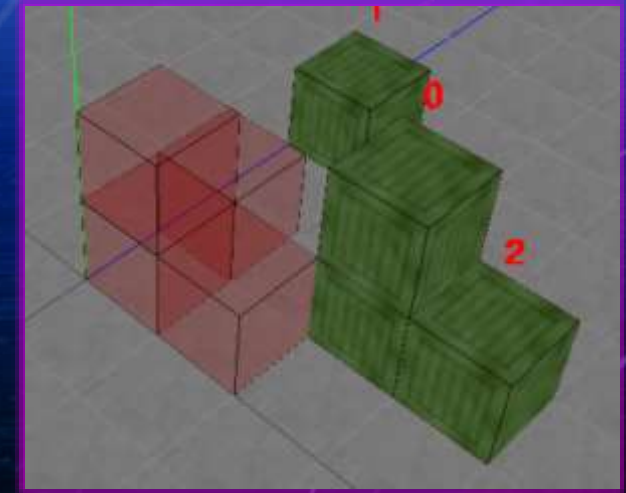
- Finds path from source to destination, **minimising collisions** with other boxes
- Moves surrounding boxes to make path **accessible**
- **Completes** all (Type 1) default scenarios



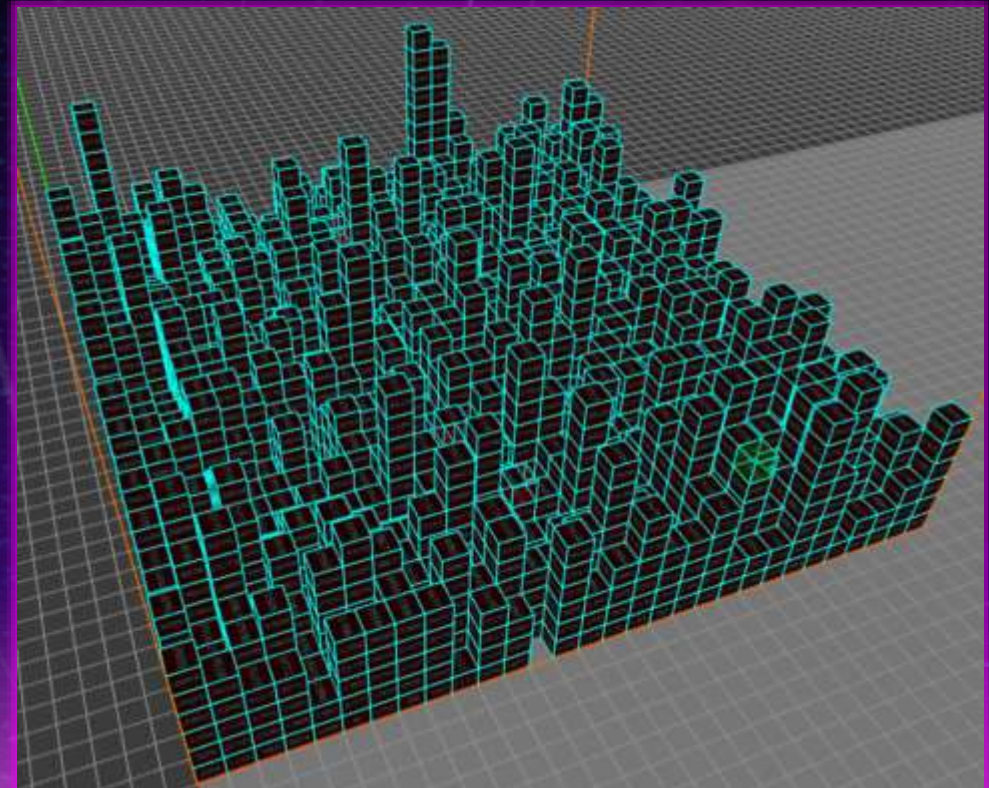
Step 2 – Automate Scenario Generation

- Created function to **automatically** generate dense scenarios for thorough **benchmarking**
- Used the function to create 10 **default scenarios**

Before



After



Step 3 – Optimise Rendering

- Used **InstancedMesh** instead of Mesh for **memory-efficiency**
- Enables very **large** scenarios (~10000 boxes) to be run **seamlessly**
- Used **anti-aliasing** for **smoother** box edges

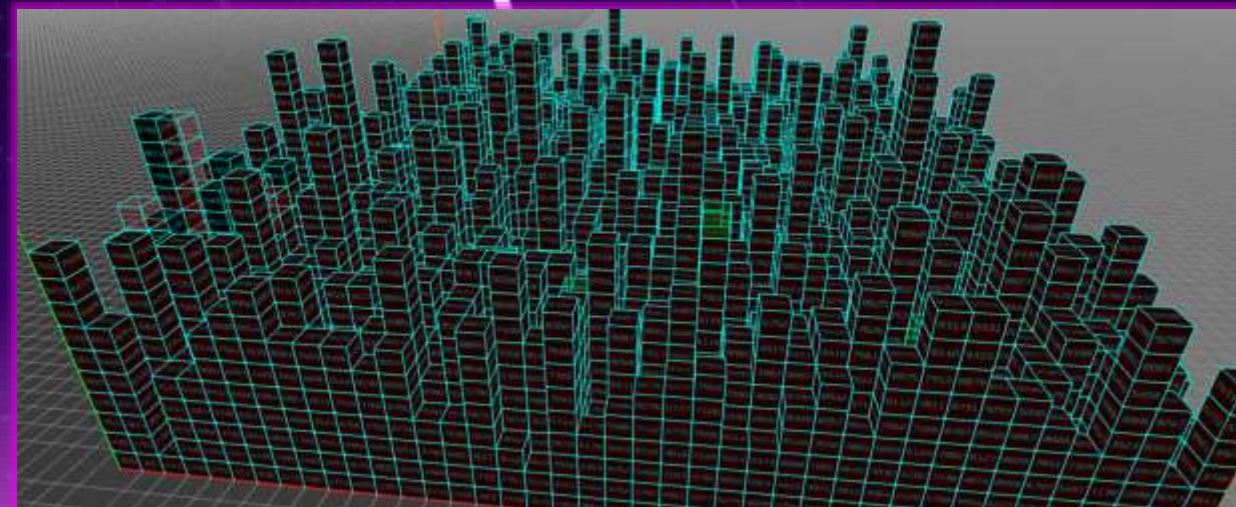
Before



After



After



Step 4 – Optimise FR Algorithm

- Used Dijkstra's to calculate **more efficient** top-down path
- Compatible moves are sent **simultaneously**
- Sped up Scenario 10 execution from **40 minutes** to **5 minutes**

Before

Scenario 10

Completed in 43m 5s

[View replay](#)

After

Scenario 10

Completed in 5m 23s

[View replay](#)

Step 5 – Improve Leaderboard

- More **intuitive** leaderboard
- Sort entries by each column
- Added **pagination** system

Before

Leaderboard 🏆

Main

This shows different sets of benchmarking across different scenarios.

Scenario: DEMO_SCENARIO Sort by: Moves 🇳🇵 SELECT

Name	Author	Datetime	Time Taken 🇳🇵	Moves 🇳🇵
Demo AS	Demo	2024-05-14 12:38:26	5	11
Kirans AS	Kiran	2024-10-09 13:07:00	5	11
Kirans AS	Kiran	2024-10-09 13:08:02	5	11

After

RoboSim Playground Instructions Leaderboard

Leaderboard

This shows different sets of benchmarking across different scenarios.

Scenario: All Scenarios Finish state: All States

Name ▲	Author ▼	Moves ▼	Time Taken ▼	Scenario ▼	Finish Time	Finish State
Random Client	Admin	6	7s	diagonal-1	2025-02-22 22:50:46	Successful
Random Client	Admin	11	12s	right-1	2025-02-22 22:50:58	Successful
Random Client	Admin	26	26s	right-10	2025-02-22 22:51:24	Failed

Step 6 – Refine Leaderboard

- Refined Leaderboard to bring back the ordering to purely better-to-worse at client's request
- Removed redundant column sorting and used new custom sorting

Before

Leaderboard

This shows different sets of benchmarking across different scenarios.

Scenario:

Finish state:

Name ▲	Author ▼	Moves ▼	Time Taken ▼	Scenario ▼	Finish Time	Finish State
Random Client	Admin	6	7s	diagonal-1	2025-02-22 22:50:46	Successful
Random Client	Admin	11	12s	right-1	2025-02-22 22:50:58	Successful
Random Client	Admin	26	26s	right-10	2025-02-22 22:51:24	Failed

After

Leaderboard

This shows different sets of benchmarking across different scenarios.

Scenario:

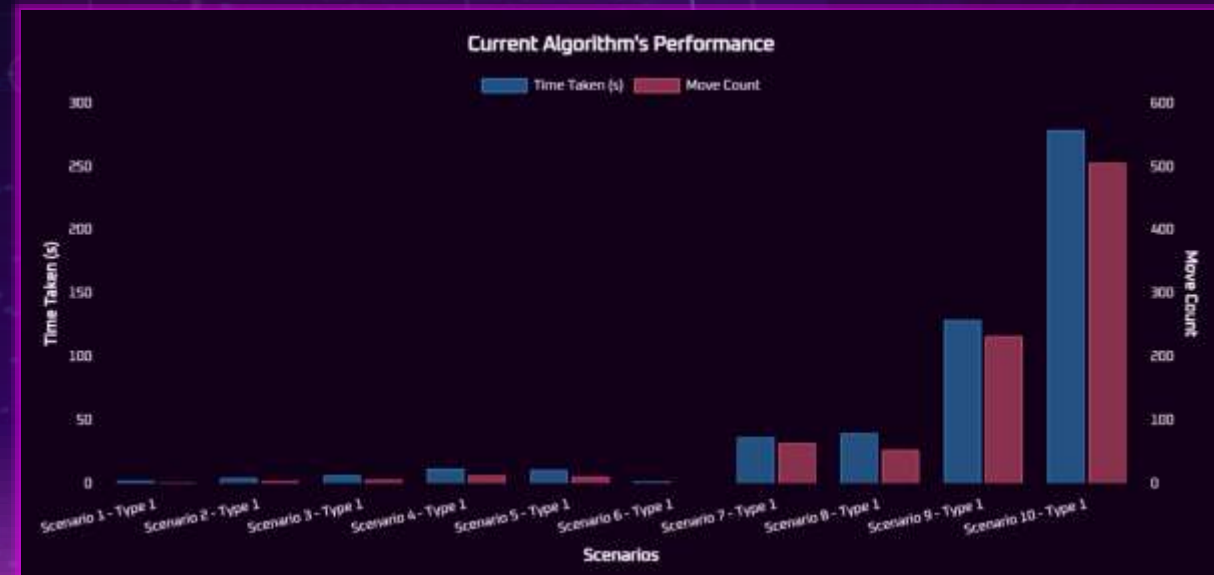
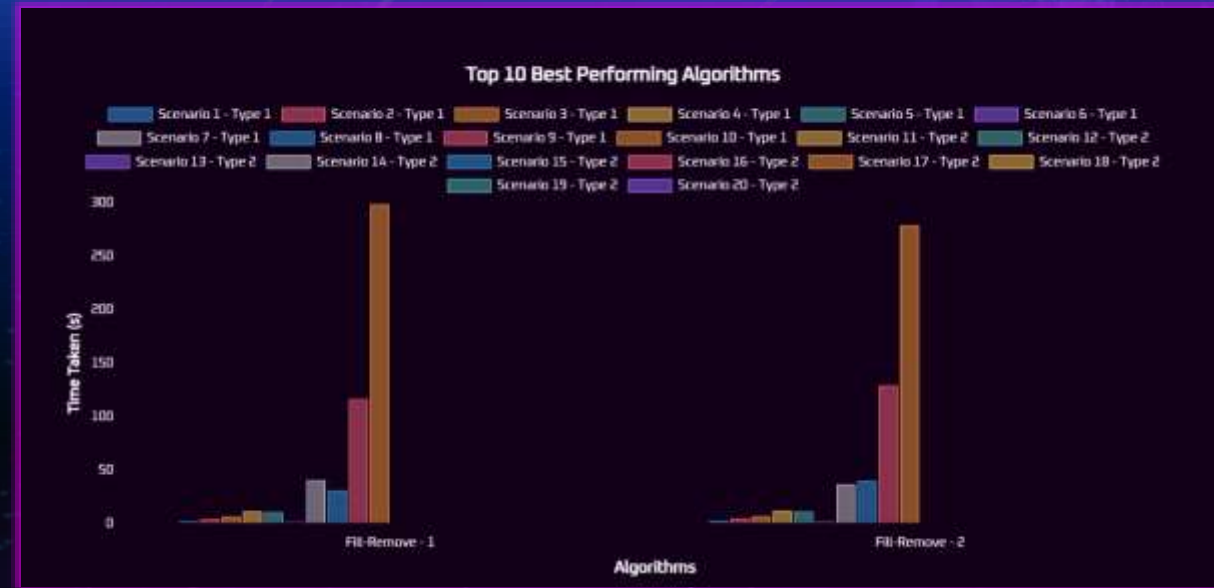
Sort:

Name	Author	Scenarios Completed	Average Moves	Average Time Taken
Ashby's Awesome Algorithm	Admin	3/10	4.0	4.8s
Ashby's Awesome Algorithm	Admin	2/10	3.0	3.5s
Ashby's Awesome Algorithm	Admin	2/10	3.0	3.8s

< 1 >

Step 7 – Create Graphing Functionality

- Created **Graphs** for in-depth **comparison** against best algorithms or in-depth **self-analysis**
- Result: Benchmarking achieved – Algorithm Designers integrated!**



Bonus Step – Marius Jurt Algorithm

- Client used our product to develop his own algorithm
- Uses A* algorithm for pathfinding
- Result: Hard proof that our product works!

```
async def algorithm(client: Client, scenario: Scenario):
    """Main algorithm to move boxes to their target locations."""
    print(f"Starting scenario: {scenario.name}")
    state = copy.deepcopy(scenario.start)
    start_set = set((cube.id, cube.x, cube.y, cube.z) for cube in state)

    for target_cube in scenario.requirements:
        if (target_cube.id, target_cube.x, target_cube.y, target_cube.z) in start_set:
            continue

        box_id = target_cube.id
        if not await clear_blocking_boxes(client, scenario, box_id, state, target_cube):
            print(f"Failed to clear blocking boxes for box {box_id}")
            await client.fail_scenario()
            return

        path = await a_star_pathfinding(client, scenario, box_id, target_cube, state)
        if not path:
            print(f"No path found for box {box_id}")
            await client.fail_scenario()
            return

        for move in path:
            try:
                pending_move = await client.send_move(move)
                await pending_move.executed()
                state.apply_move(move)
                print(f"Executed move for box {box_id}: {move._dict_}")
            except Exception as e:
                print(f"Move failed: {e}")
```

Algorithm Client Summary

Name: A* - mj Author: Marius State: Failed

Scenario	State	View
Scenario 1	Completed in 2.4s	View replay
Scenario 2	Completed in 4.6s	View replay
Scenario 3	Completed in 7.0s	View replay
Scenario 4	Completed in 16.8s	View replay
Scenario 5	Completed in 11.4s	View replay
Scenario 6	Completed in 1.6s	View replay
Scenario 7	Failed after 4m 37s	View replay

IDP 3: Authentication & Security

Authentication & Configurability

- Account-based system
- Users receive token on registration to run algorithms
- Account page to modify details
- Result: Security & configurability integrated!

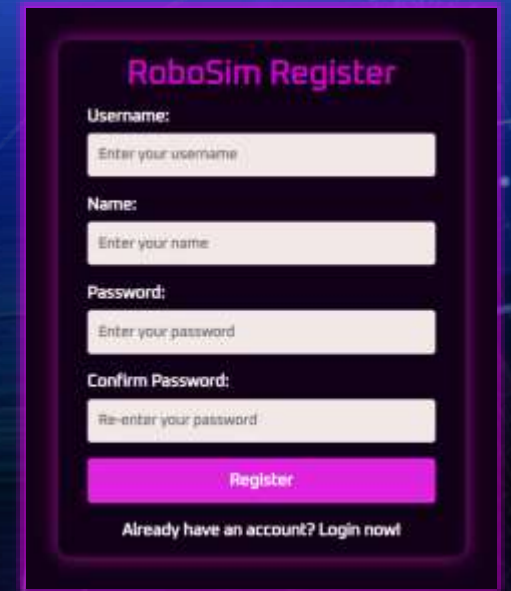


RoboSim Login

Username:

Password:

Do not have an account? Register now!



RoboSim Register

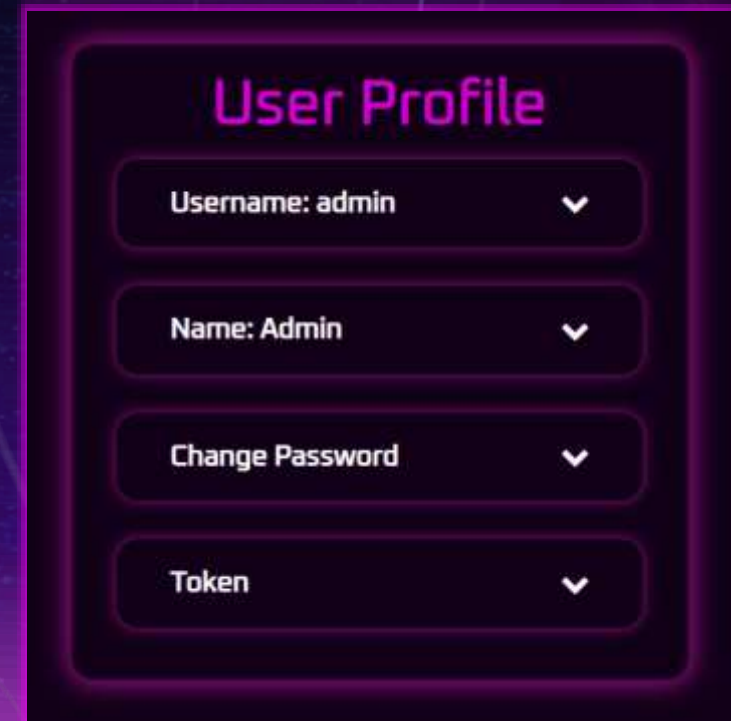
Username:

Name:

Password:

Confirm Password:

Already have an account? Login now!



User Profile

Username: admin ▼

Name: Admin ▼

Change Password ▼

Token ▼

IDP 4: Playground & User Testing

Step 1 – Improve Box Controls

- Removed **redundant** cardinal buttons
- Box movement **solely** controlled via keypresses

Before

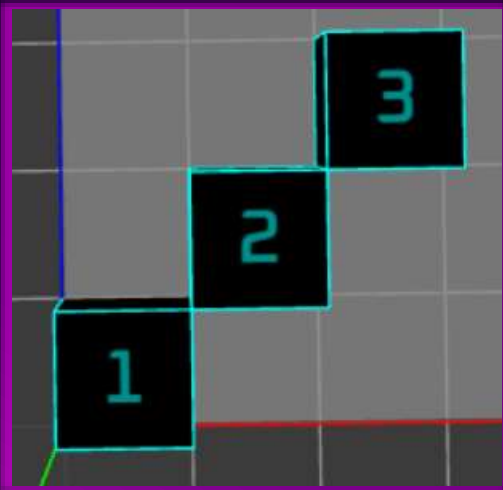


After



User Testing – Demo Scenes

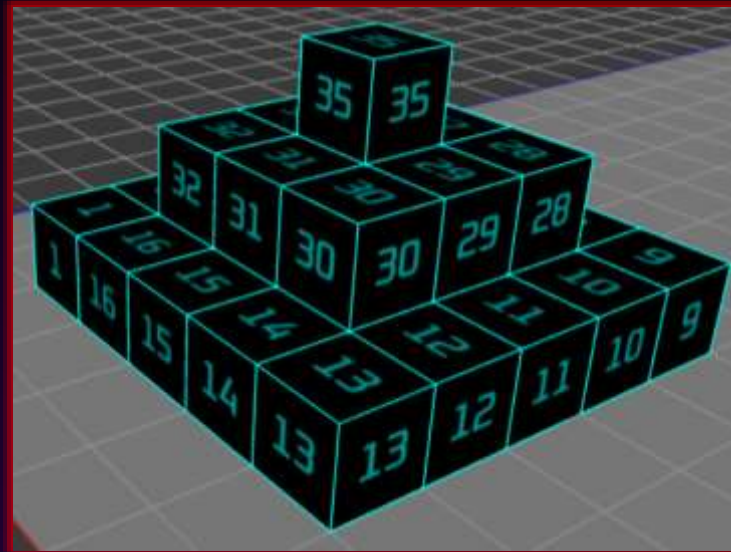
Scene 1



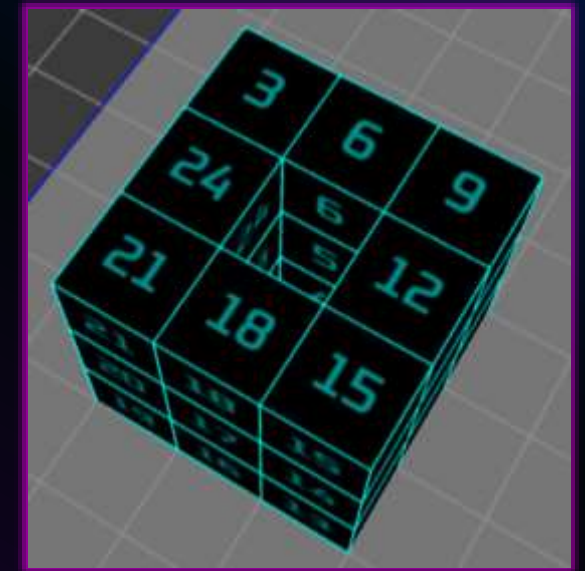
Scene 2



Scene 3



Scene 4



User Testing 1 – Quantitative

NASA-TLX



SUS



Mental Demand



Performance



Effort



Frustration



User Testing 1 – Qualitative

- Positives:
 - Fast rendering of boxes
 - Speedy movement of boxes
- Negatives:
 - Inconvenient to have to manually manage ID
 - Difficult to stack boxes
 - Box movement restrictions unexplained
 - Select Box feature not working

Step 2 – Automate ID Management

- Abstracted ID management from user
- Added **keypresses** for box creation & deletion
- Added **box restrictions** to Instructions

Before

The 'Before' UI state shows a control panel with the following elements:

- Buttons: 'Box ID (Default: 0)', 'Add Box', 'Remove Box', 'Clear Scene' (red), 'Save Scene' (green), 'Choose file No fil...osen', 'Load Scene' (yellow), 'X Size', 'Y Size', 'Z Size', 'Scenario Name', 'Set Start', 'Set End', 'Download Scenario'.

After

The 'After' UI state shows a control panel with the following elements:

- Buttons: 'Selected Box: None', 'Add Box', 'Remove Box', 'Clear Scene' (red), 'Save Scene' (green), 'Choose file No fil...osen', 'Load Scene' (yellow), 'X Size', 'Y Size', 'Z Size', 'Scenario Name', 'Set Start', 'Set End', 'Download Scenario', 'Reset State'.

User Testing 2 – Qualitative

- - Positives:
 - Automatic management of IDs
 - Engagement with all scenes
 - Speedy movement of boxes
 - Negatives:
 - Difficult to stack boxes
 - Complex structures time-consuming to build
- Unanimous support for Forklift Mode (enables users to move stacks of boxes)

Step 3 – Forklift Mode

- Added **Forklift Mode** as a **toggle** feature (best compromise)
- **Styled** controls menu to with new colour scheme
- **Simplified** Load Scene functionality

Before

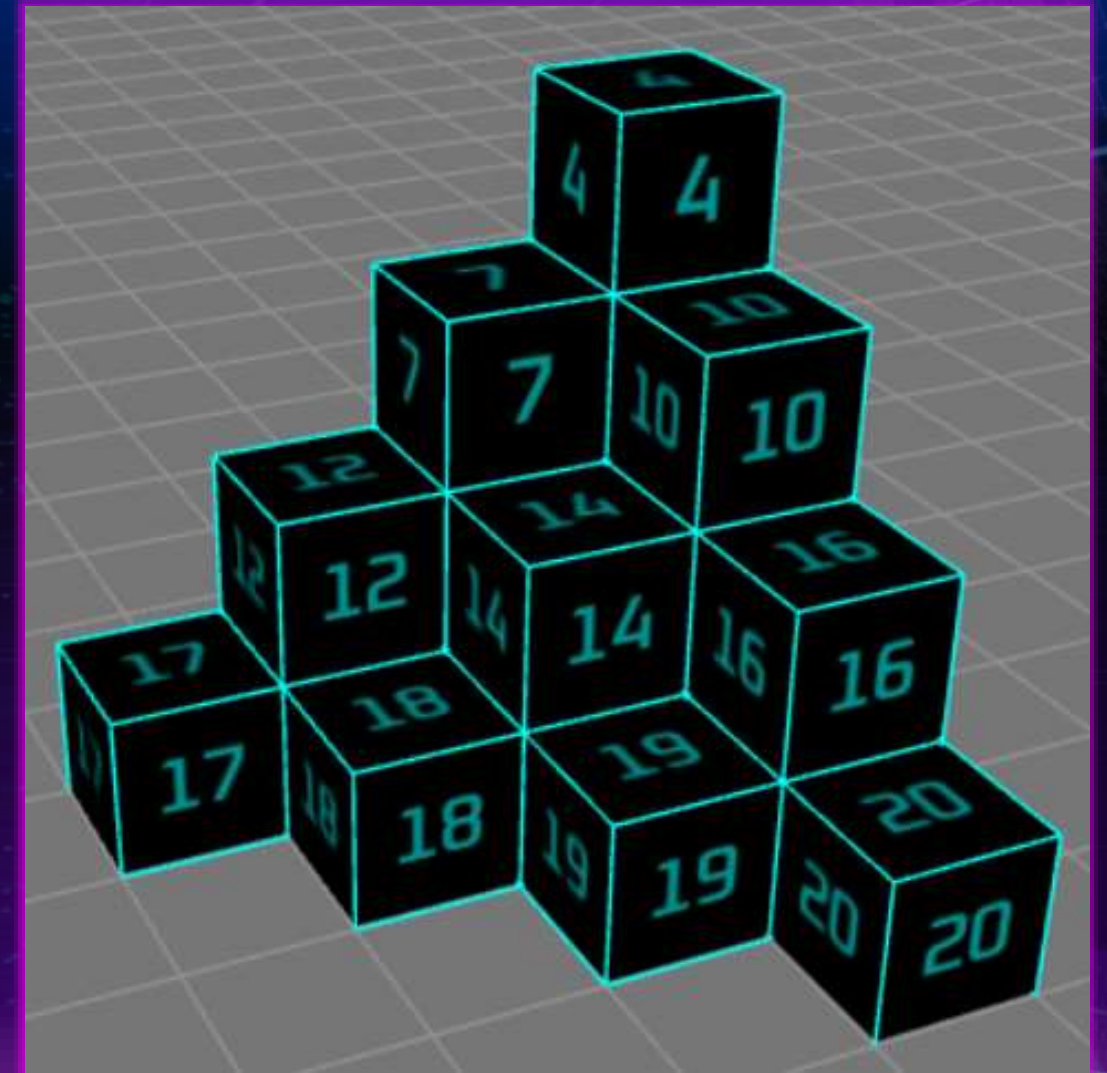


After



Bonus Scene 5

- Created in anticipation of users completing scenes faster
- Only allowed to do if they completed **all** other scenes



User Testing 3 – Quantitative

NASA-TLX



SUS



Mental Demand



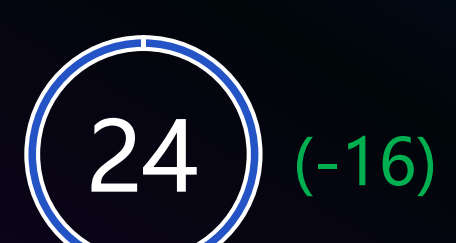
Performance



Effort



Frustration



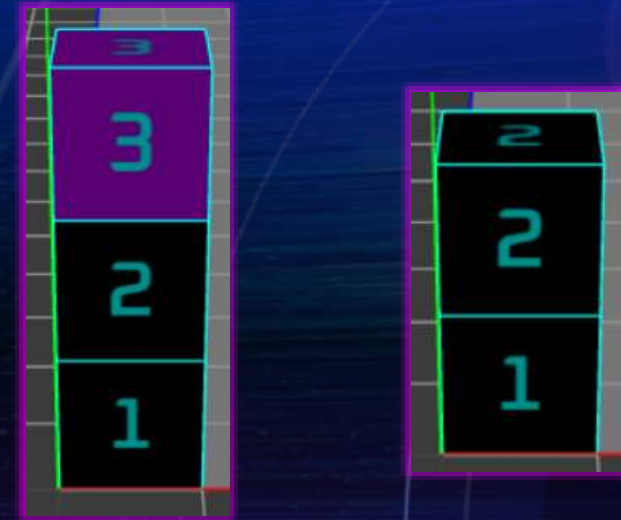
User Testing 3 – Qualitative

- Positives:
 - Style of warehouse boxes
 - Forklift Mode (can create structures much faster)
 - Completion of all scenes (including Bonus Scene)
- Negatives:
 - Selected box resets to None after every box removal
 - Hard to navigate boxes when camera is not at default angle
 - Cannot easily see which box is selected
 - Limited indication of invalid actions

Step 4 – Improve Box Controls Further

- **Stack-based** Remove Box functionality for **easier** box deletion
- **Camera-based** box movement directions for **easier** box navigation

Before



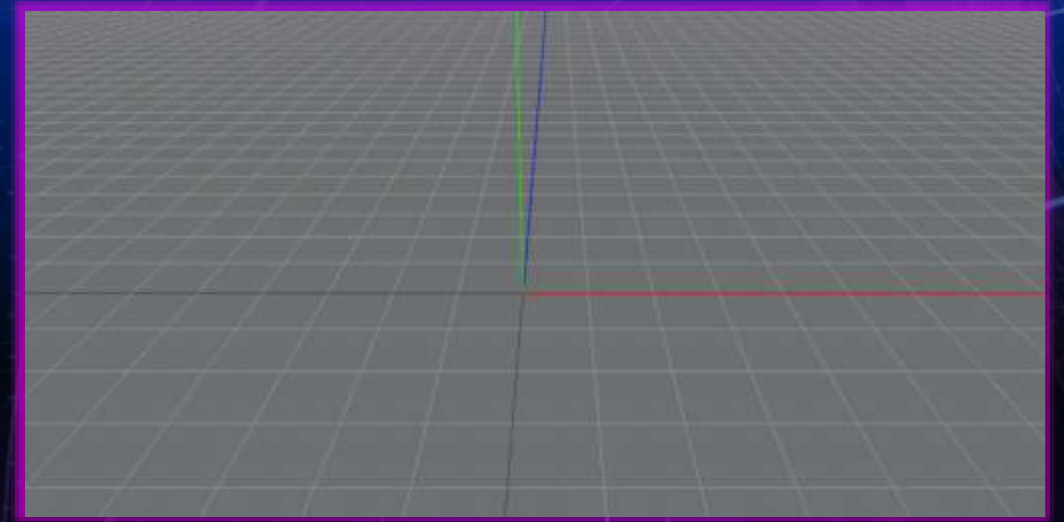
After



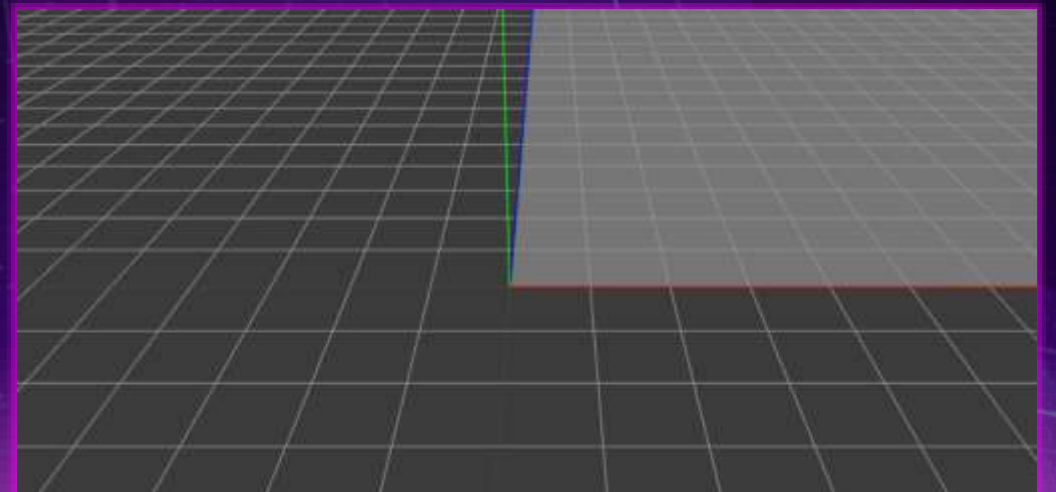
Step 5 – Clearer Warehouse Boundaries & Validation

- Made warehouse boundaries clearer
- Playground pop-up errors for invalid actions
- Result: Warehouse Managers & Algorithm Designers integrated!

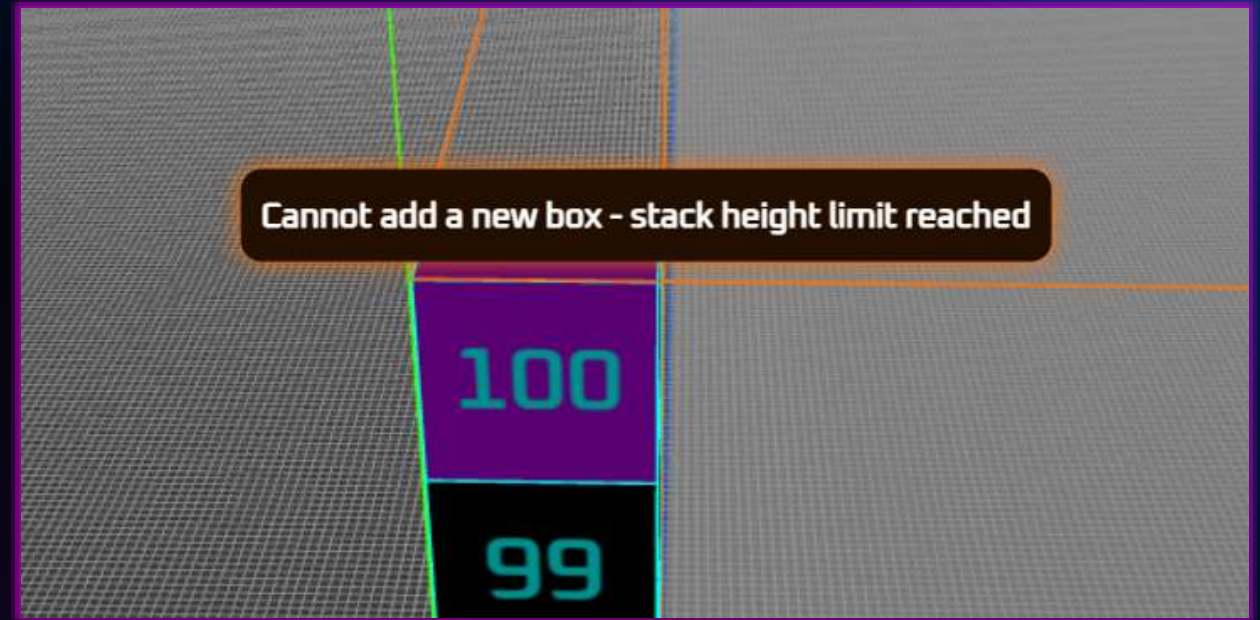
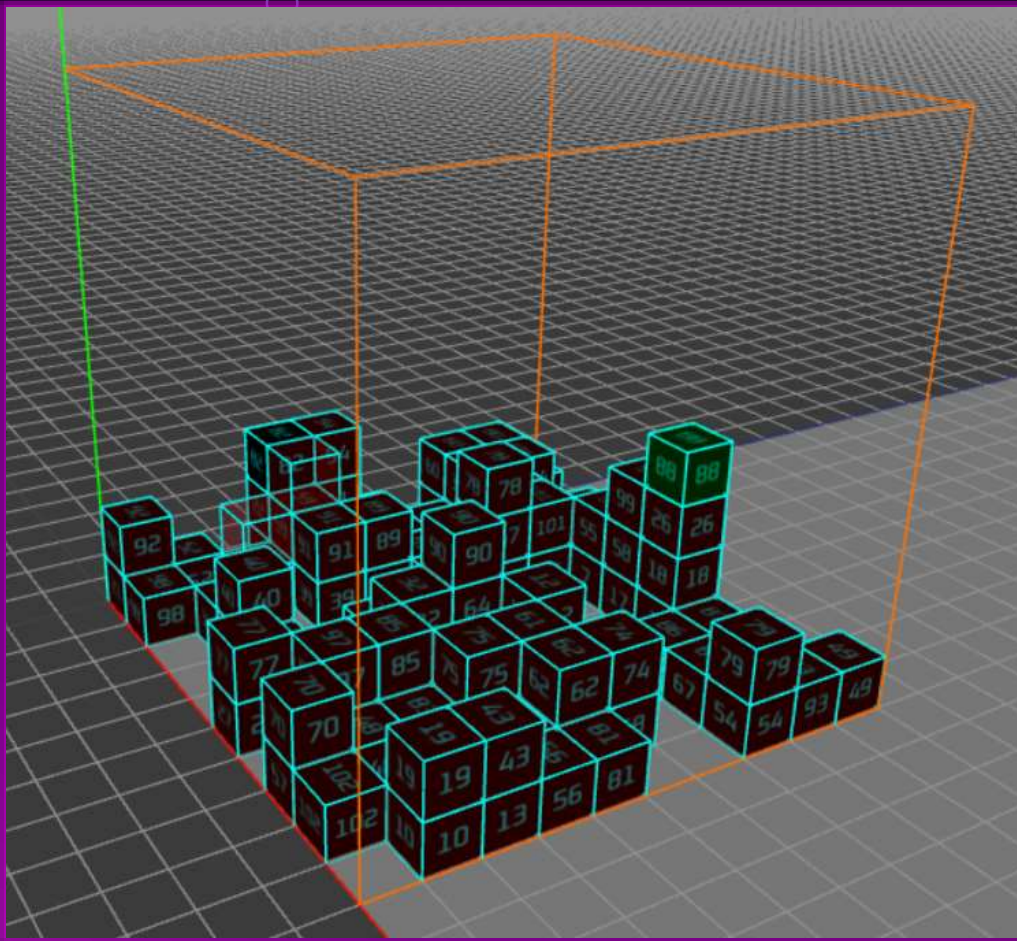
Before



After



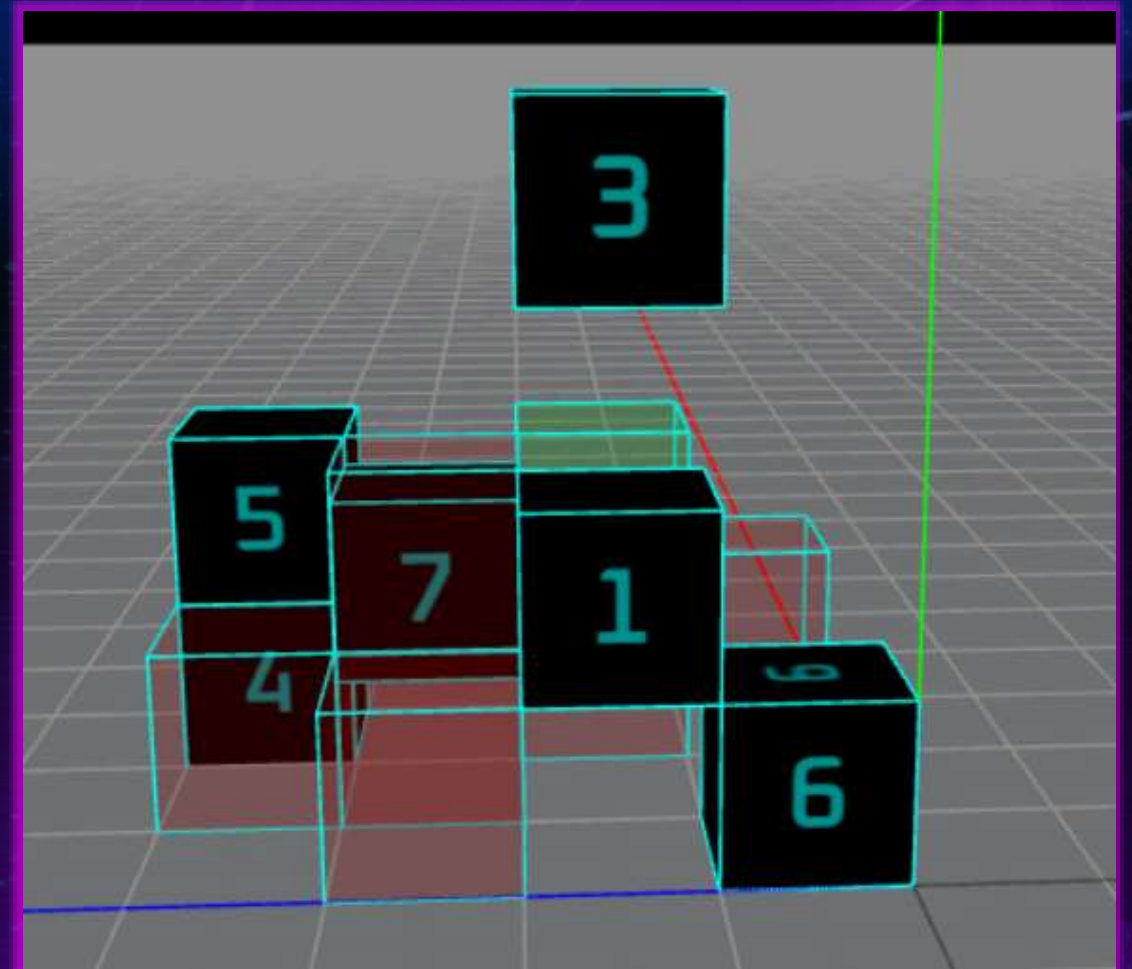
Step 5 – Clearer Warehouse Boundaries & Validation



IDP 5: Second Box Type

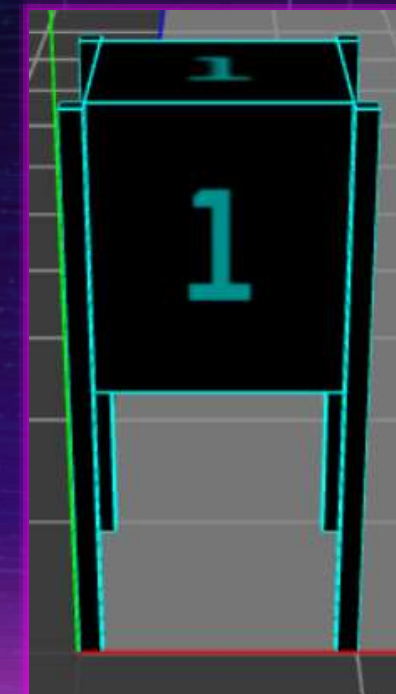
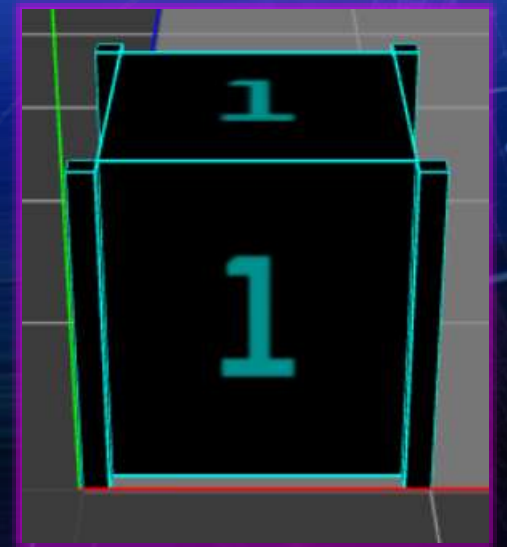
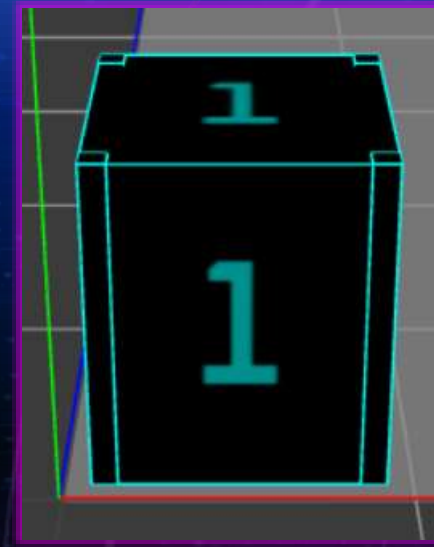
Step 1 – Type 2 Box Scenarios

- Made **sweeping changes** to the server (again) to accommodate this new box type
- Created 10 Type 2 **default scenarios** for the Type 2 Boxes



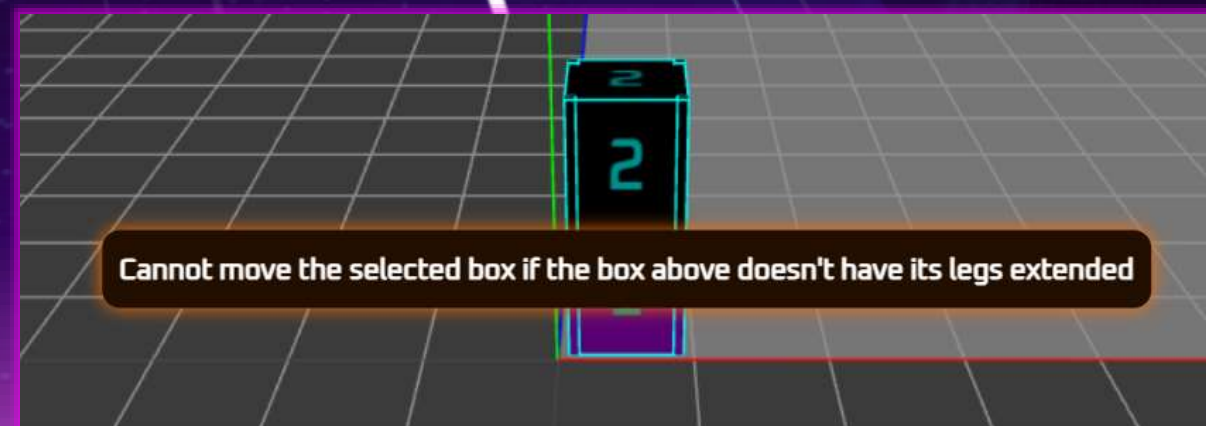
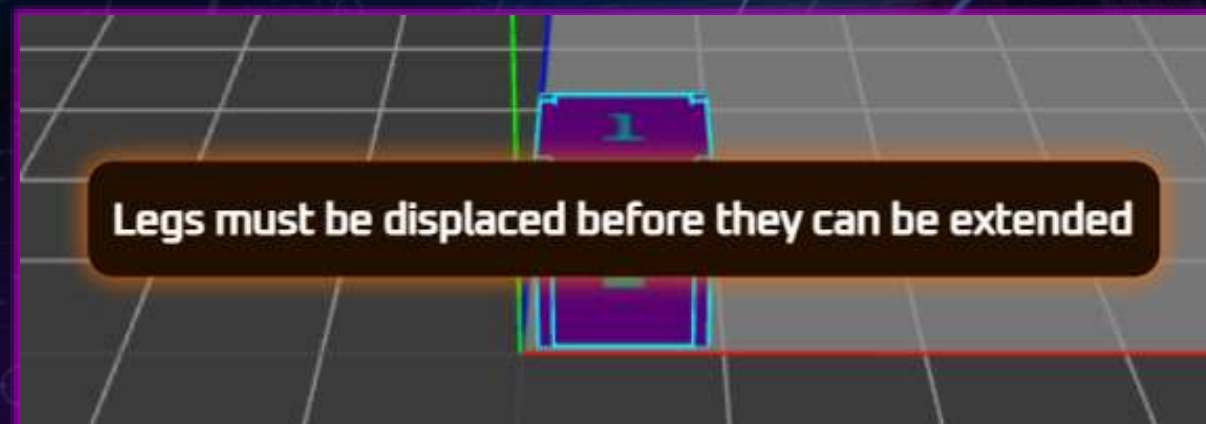
Step 2 – Type 2 Box Animations

- Added **visual** robot legs to Type 2 boxes to **differentiate** box types
- Created **animations** for extension/retraction of box legs



Step 3 – Type 2 Box Playground

- Created **full** Playground functionalities for Type 2 boxes
- Type 2 pop-up errors to teach users about **new mechanics**
- **Result: Toshiba Developers integrated!**



Demonstration (New Project)

Testing & Project Process

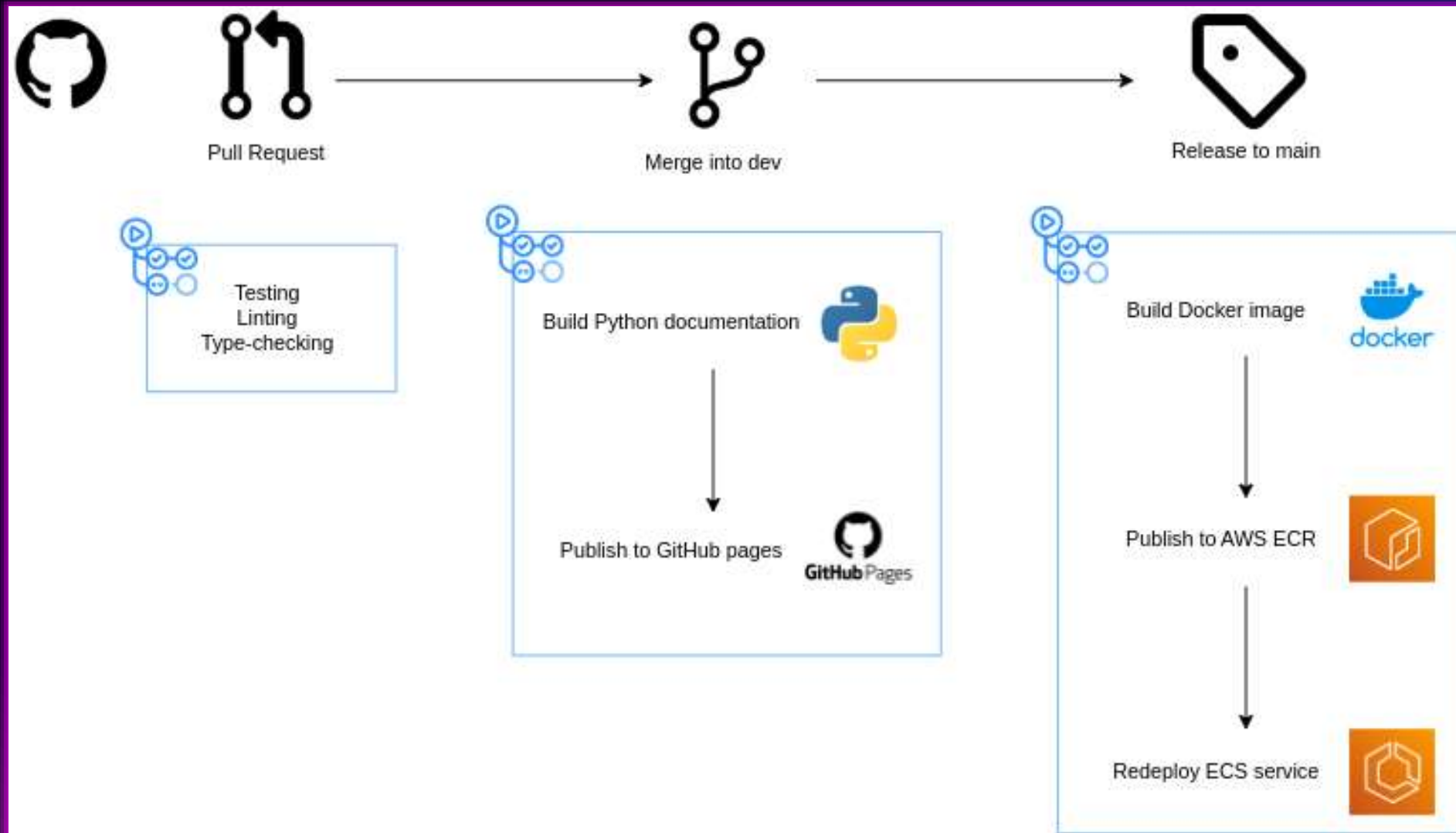
Testing & Deployment

- 124 tests on the server
- 38 Python tests
- 18 end-to-end tests
- 91% test coverage on the server

```
===== Coverage summary =====
Statements   : 91.47% ( 3790/4143 )
Branches     : 81.8% ( 454/555 )
Functions    : 96.81% ( 152/157 )
Lines        : 91.47% ( 3790/4143 )
=====
```

src		94.29%	794/842	78.7%	85/108	96.07%	49/51	94.29%	794/842
src/api		81.73%	470/575	75.86%	44/58	100%	0/0	81.73%	470/575
src/connect		86.67%	898/1036	79.36%	100/126	95.34%	41/43	86.67%	898/1036
src/model		96.2%	1444/1501	85.76%	223/260	98.41%	62/63	96.2%	1444/1501
types		100%	180/180	100%	5/5	100%	0/0	100%	180/180

CI/CD



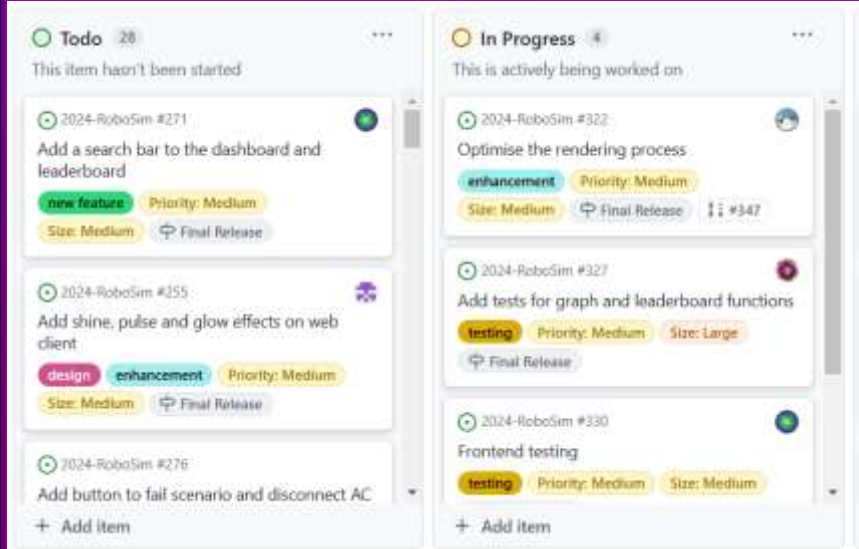
Project Management (Part 1)

- Frequent meetings with client
- Full metadata for issues & PRs
- In-depth code reviews for a mature process

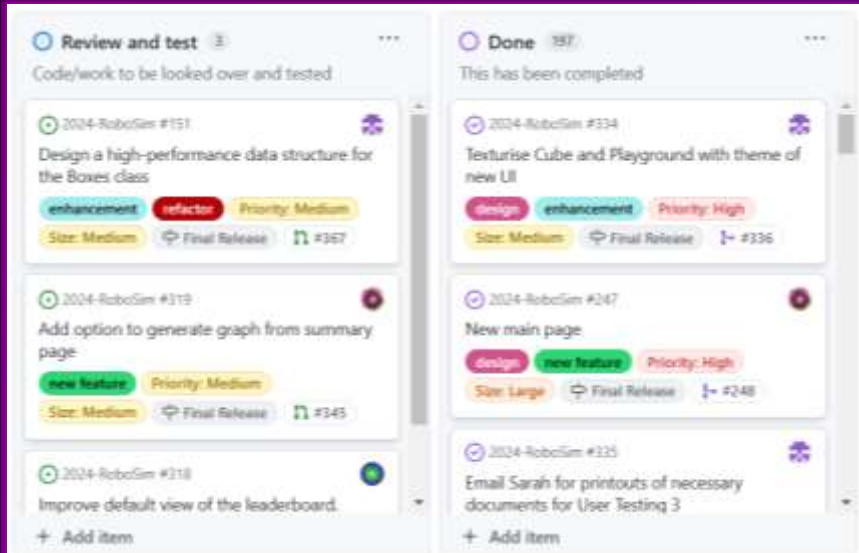
☐  **Graph generation** ✓ new feature 1  58
#281 by si23870 was merged on Mar 5 • Approved  Final Release

☐  **feat: User login system** ✓ new feature 1  54
#189 by DanielZhang05 was merged last month • Approved  Beta Release

☐  **feat: automate scenario generation** ✓ new feature testing 1  29
#243 by davidw1281 was merged 2 weeks ago • Approved  Beta Release



The image shows a Kanban board with two columns: 'Todo' and 'In Progress'. The 'Todo' column has three items: '2024-RoboSim #271' (Add a search bar to the dashboard and leaderboard, new feature, Priority: Medium, Size: Medium, Final Release), '2024-RoboSim #355' (Add shine, pulse and glow effects on web client, design, enhancement, Priority: Medium, Size: Medium, Final Release), and '2024-RoboSim #276' (Add button to fail scenario and disconnect AC, testing, Priority: Medium, Size: Medium, Final Release). The 'In Progress' column has three items: '2024-RoboSim #322' (Optimise the rendering process, enhancement, Priority: Medium, Size: Medium, Final Release, #347), '2024-RoboSim #327' (Add tests for graph and leaderboard functions, testing, Priority: Medium, Size: Large, Final Release), and '2024-RoboSim #330' (Frontend testing, testing, Priority: Medium, Size: Medium, Final Release). Each item has a status icon (green circle for Todo, yellow circle for In Progress) and a 'Final Release' button.



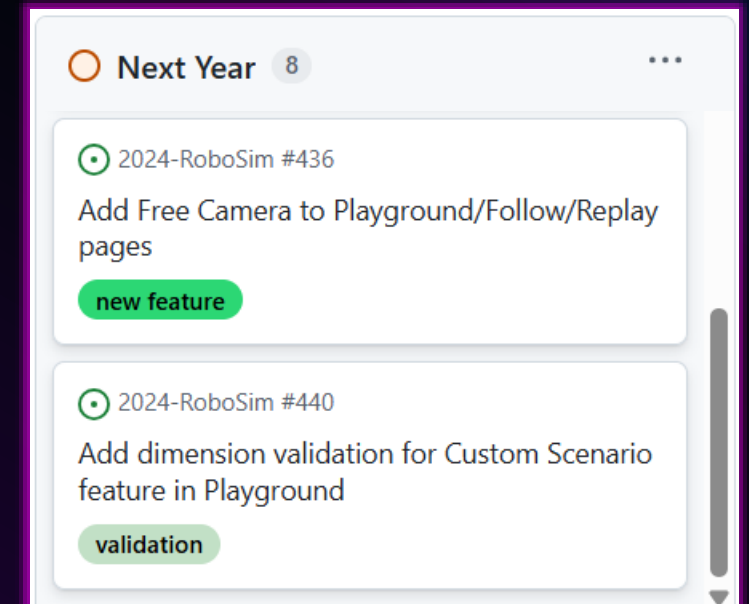
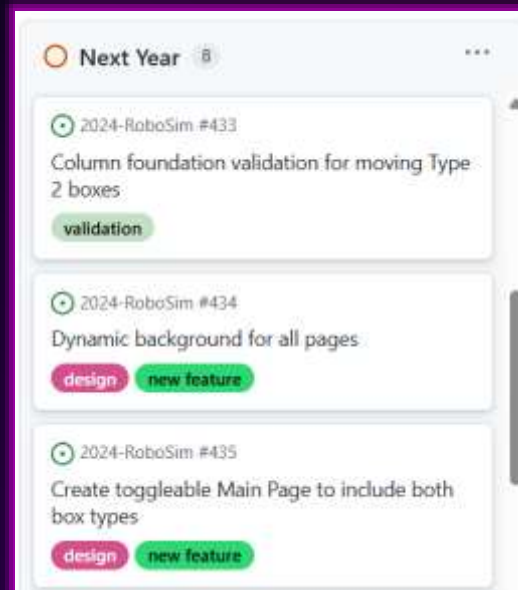
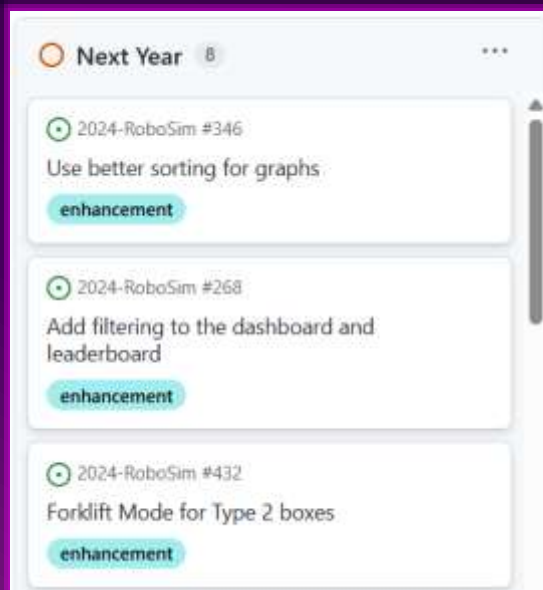
The image shows a Kanban board with two columns: 'Review and test' and 'Done'. The 'Review and test' column has three items: '2024-RoboSim #151' (Design a high-performance data structure for the Boxes class, enhancement, refactor, Priority: Medium, Size: Medium, Final Release, #367), '2024-RoboSim #319' (Add option to generate graph from summary page, new feature, Priority: Medium, Size: Medium, Final Release, #345), and '2024-RoboSim #318' (Improve default view of the leaderboard, testing, Priority: Medium, Size: Medium, Final Release). The 'Done' column has three items: '2024-RoboSim #334' (Texturise Cube and Playground with theme of new UI, design, enhancement, Priority: High, Size: Medium, Final Release, #336), '2024-RoboSim #247' (New main page, design, new feature, Priority: High, Size: Large, Final Release, #248), and '2024-RoboSim #325' (Email Sarah for printouts of necessary documents for User Testing 3, testing, Priority: Medium, Size: Medium, Final Release). Each item has a status icon (blue circle for Review and test, purple circle for Done) and a 'Final Release' button.

Project Management (Part 2)

- - Utilised more **channels** of communication (**WhatsApp Groupchat**)
 - Removed **redundant** Monday meetings
 - Adopted **AGILE methodology** with **weekly sprints**
 - Typical **Weekly Sprint**:
 - Wednesday: **New Sprint** Begins
 - Thursday – Saturday: **Coding**
 - Sunday: **Rest** – minimal activity
 - Monday – Tuesday: **Code Reviews**

Project Management (Part 3)

- **Delegation of tasks** from more experienced members
- **NO MORE SQUASHING COMMITS!!!!**
- Gave **starter tasks** for **next iteration** to client (and put them on **Kanban Board**)



Project Process Log

- The journey, in full
- Explains project-based decisions to future iterations
- Encourages proper documentation practices among group
- Easy to linearise at end of project
- Allows intangible workload to be recognised

Problems & Solutions



Problems

Operating as a **team of 3** instead of 5

Losing a team member

More group **disagreements**

Used over **half** of entire monthly budget for **GitHub Actions**

Solutions

Kept the door open – teammate **came back** into the group

Increased efficiency within team to **mitigate negative impact** and **maintain** work rate

Considered multiple angles to come to a **holistic** solution that **maximised** benefit for our **IDP**

Made CI more **efficient**

Project Reflection



What We Learned

Algorithm Design

Working with legacy code

Documentation skills

UI Design

Iterative Design

Process Maturity & Repeatability

What We Would Do Differently

Use of AGILE from the start

More productive meetings in TB1

More engagement in TB1

Getting to grips with legacy code faster

Test-Driven Development

Thank You